

Applications of the Delta Method

A concise collection of important applications of the Delta Method in asymptotic statistics.

1. Sample Standard Deviation

Suppose $\sqrt{n}(S_n - \sigma) \Rightarrow N(0, \tau^2)$. Taking $g(x) = \sqrt{x}$ gives $g'(\sigma^2) = 1/(2\sigma)$. Hence $\sqrt{n}(S_n - \sigma) \Rightarrow N(0, \tau^2/(4\sigma^2))$.

2. Logarithm of the Sample Mean

If $\sqrt{n}(X_n - \mu) \Rightarrow N(0, \sigma^2)$, then with $g(x) = \log x$, $\sqrt{n}(\log X_n - \log \mu) \Rightarrow N(0, \sigma^2/\mu^2)$.

3. Reciprocal Estimators

If $\sqrt{n}(T_n - \theta) \Rightarrow N(0, \sigma^2)$ and $\theta \neq 0$, then $g(x) = 1/x$ yields $\sqrt{n}(1/T_n - 1/\theta) \Rightarrow N(0, \sigma^2/\theta^2)$.

4. Odds Estimators

For $g(p) = p/(1-p)$, $g'(p) = 1/(1-p)^2$. Apply the Delta Method to obtain the asymptotic distribution of estimated odds.

5. Confidence Intervals for Nonlinear Parameters

If $\psi = e^\theta$ and T_n estimates θ , then $g(x) = e^x$ implies $\sqrt{n}(e^{T_n} - e^\theta) \Rightarrow N(0, e^{2\theta}\sigma^2)$.

6. Fisher's z-Transformation

For $g(\rho) = \frac{1}{2} \log((1+\rho)/(1-\rho))$, the derivative is $g'(\rho) = 1/(1-\rho^2)$. This gives the asymptotic variance of transformed correlations.

7. Coefficient of Variation

Using the multivariate Delta Method with $g(\mu, \sigma) = \sigma/\mu$, derive the asymptotic variance of the coefficient of variation.

8. Logistic Regression Odds Ratios

If β is asymptotically normal, then e^{β} is analyzed via $g(x) = e^x$. The approximate variance becomes $e^{2\beta} \text{Var}(\beta)$.

9. Multivariate Delta Method

If $\sqrt{n}(T_n - \theta) \Rightarrow N(0, \Sigma)$, then $\sqrt{n}(g(T_n) - g(\theta)) \Rightarrow N(0, \nabla g(\theta)^T \Sigma \nabla g(\theta))$.

10. Second-Order Delta Method

When $g'(\theta) = 0$, a second-order Taylor expansion is required. Example: $g(x) = x^2$ at $\theta = 0$ leads to non-normal limits.

Common Transformations

$\log(x)$: derivative = $1/x$ $1/x$: derivative = $-1/x^2$ $\sqrt{x}$: derivative = $1/(2\sqrt{x})$ e^x : derivative = e^x $x/(1-x)$:
derivative = $1/(1-x)^2$ $\log(x/(1-x))$: derivative = $1/[x(1-x)]$